



Software Testing and Reliability

Module I : Fundamentals of Testing

1. Human and errors, Defects,
2. Faults, Failures,
3. Defect Rate and Reliability,
4. Defect Prevention,
5. Reduction and Containment
6. Testing and Debugging
7. Software Quality
8. Requirement Behavior and Correctness
9. Fundamentals of Test Process
10. Psychology of Testing
11. General Principles of Testing
12. Test Metrics
13. Agile Methodology and Its Impact on testing
14. Test Levels: Unit, Component

15. Module, Integration, System, Acceptance, Generic

Module II: Approaches to Testing

1. Static Testing
2. Structured Group Examinations
3. Static Analysis
4. Control flow & Data flow
5. Determining Metrics
6. Dynamic Testing
7. Black Box Testing: Equivalence Class Partitioning
8. Boundary Value Analysis
9. State Transition Test,
10. Cause Effect Graphing and Decision Table Technique and Used Case Testing and Advanced black box techniques
11. White Box Testing: Statement Coverage, Branch Coverage,
12. Path Coverage, System integration,
13. Deployment testing,
14. Beta testing, Scalability testing,
15. Reliability testing,
16. Stress testing,
17. Acceptance testing: Acceptance criteria, test cases selection and execution

Module III : Test Management

1. Test Organization,

2. Test teams, tasks and Qualifications
 3. Test Planning,
 4. Quality Assurance Plan
 5. Test Plan, Prioritization Plan
 6. Test Exit Criteria
 7. Cost and economy Aspects
 8. Test Strategies
 9. Preventive versus Reactive Approach
 10. Analytical versus heuristic Approach
 11. Test Activity Management
 12. Incident Management,
 13. Configuration Management,
 14. Test Progress Monitoring and Control
 15. Specialized Testing: Performance, Load, Stress & Security Testing
-

Module IV : Software Reliability

1. Defining Software Reliability
2. Software Reliability Attributes and Specification
3. Concept of Introduction to Measurement and Inspection Process
4. Documents and Metrics
5. Basics of Reliability Theory
6. Software Reliability Problem
7. Modeling Process
8. Software Reliability Models (SRGM)

9. preliminary Concepts of Reliability Engineering
 10. Parameter Estimation
 11. Model Validation
 12. Execution Time Models
 13. Calendar Time Models
 14. Erlang Model
 15. Modeling Fault Dependency and Debugging Time Lag
 16. Testing Effort Dependent Modeling
 17. Distributed Environment
 18. Imperfect Debugging
 19. Testing-Domain Models
 20. Change-Point
 21. Unified Approach Concepts
 22. Artificial Neural Networks based SRGM
 23. Introduction to Discrete SRGM.
-

Module V : Software Reliability Allocation Models

1. Software Reliability Allocation Models
2. Criteria for Model Evaluation
3. Optimal Reliability Allocation
4. Quality Planning and Control
5. Quality Improvement Process
6. Evolution of Software Quality Assurance (SQA)
7. Major SQA Activities
8. Major SQA Issues

- 9. Zero Defect Software
- 10. Verification and Validation
- 11. Evolutionary Nature of Verification and Validation,
- 12. Impracticality of Testing all Data and Paths
- 13. Proof of Correctness
- 14. Software Testing
- 15. Functional
- 16. Structural and Error-Oriented Analysis & Testing,
- 17. Static and Dynamic Testing Tools,
- 18. Characteristics of Modern Testing ToolsSelf-Work/ Professional Skill
Development activities